



DUST DEVIL

Climate change may be driving an expansion of Valley fever, a deadly fungal infection spread by airborne spores *By Meredith Wadman*

In the spring of 2021, shortly after moving from Seattle to Arizona, Marc Evans, a retired broadcast industry lawyer, began tending the vegetable garden in his niece's backyard. Evans and his wife were staying in a guest house on the property, in the Phoenix suburb of Paradise Valley, while they house hunted. As Evans, then 71, turned soil to plant tomatoes, a backhoe was cratering the neighbor's yard next door to put in a swimming pool.

A few weeks later, Evans began waking up at night short of breath. As his symptoms got worse, he started creeping out to the couch to sleep sitting up, so as not to disturb his wife, Arlene Evans. She already worried about his health: He is diabetic and had weathered two heart attacks decades earlier.

"I tend to just sort of ride [illnesses] out until they go away," Marc says. "Well, this one didn't go away. This one got worse, and

I didn't want her to worry about it, so I let it go on longer than I should."

At night, he started seeing floating black-and-white checkerboards and stripes that he knew weren't there. "That was some scary shit," he recalls. He broke down and told Arlene, who took him to the emergency room (ER) in the middle of the night.

At the nearby Mayo Clinic Hospital, he was soon diagnosed with Valley fever, or coccidioidomycosis—a respiratory disease contracted by inhaling microscopic spores from the soil-dwelling fungus *Coccidioides*. Marc had never heard of the fungus, endemic in Arizona and California, whose San Joaquin Valley gives the disease its name. "You are a resident for a long time here, you know it's out there," he says. "But people who move from other places? They don't have a clue."

"Cocci" (pronounced "coxei")—a nickname used, imprecisely, for both the fun-

gus and the disease it causes—had taken up residence in Marc's left lung. There, an opaque, baseball-size blot was visible on his imaging results. Doctors stuck a needle in it and withdrew cells and fluid laced with the fungus. Evans also had hints of meningitis, an inflammation of the membranes that envelop the brain and spinal cord. Beyond the floating checkerboards, he was complaining of headaches. But tests of his spinal fluid came back frustratingly ambiguous. He was started on high-dose antifungal medication and went home after 7 days.

Weeks later, Arlene rushed him back to the ER. His skin was gray and his shortness of breath had returned with a vengeance. Minutes after Marc was whisked away, three masked doctors emerged to ask Arlene whether he had a do not resuscitate order.

"I'm like, 'What are you talking about here?'" she remembers. "They told me all of his organs had failed."

Workers harvest garlic in dusty conditions near Bakersfield, California. Soil-roiling work can stir up the fungal spores that cause Valley fever.

wet confines of the lung, morph into structures called spherules that burst to release boatloads of tiny endospores that become new spherules, continuing the cycle. Most of these people have a flulike illness lasting weeks or months. But 5% to 10% of cases result in lifelong lung infections, sometimes forcing people to be on powerful antifungal medications permanently. In about 1.5% of illnesses, the fungus disseminates, attacking bones, joints, or skin, or causing meningitis. Invasive disease is more common in men, a skew not entirely explained by higher occupational exposures. Black people, like Marc, are at strikingly higher risk, for unknown reasons.

And cases are escalating fast. Diagnoses reported to the U.S. Centers for Disease Control and Prevention (CDC) have ballooned from about 2800 annually at the turn of the century to about 20,000 in 2023, with at least 200 people dying each year. Arizona and California, where roughly 97% of U.S. cases are reported, have seen dramatic increases recently: Incidence in Arizona has grown by 73% in the past 10 years—to 146 cases per 100,000 people. In California, incidence quadrupled between 2014 and 2023 to 23.2 cases per 100,000 people.

“We’re just up to our neck on the clinical side,” says Carlos D’Assumpcao, an infectious disease physician at the Valley Fever Institute, part of Kern Medical in Bakersfield, California, which has hired two new doctors since 2018 to keep up. In the 12 months that ended in November 2024, the institute’s doctors saw Valley fever patients nearly 6500 times.

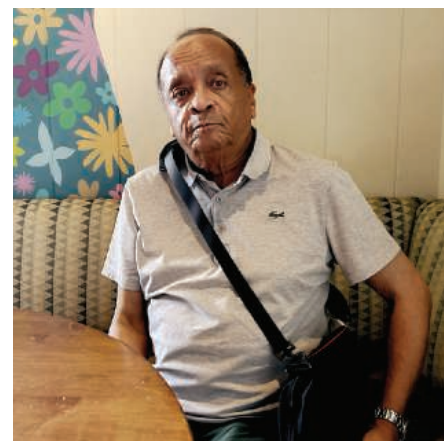
Changes in reporting and speedier, more sensitive diagnostic tests may have contributed to the upward case trend. But Mitsuru Toda, an epidemiologist who is CDC’s endemic fungal diseases expert, says it likely also represents a real increase in illnesses. For one thing, the population of the Southwest has shot up, exposing thousands of new residents without preexisting immunity to the fungus. (In the 12 months before Marc came down with Valley fever, Phoenix added 13,000 people.) And *Coccidioides* is spreading geographically. In several counties of both Arizona and California where cases had been relatively rare, disease incidence has increased dramatically during the past 5 to 10 years.

One likely contributor is a warming climate. The fungus thrives in hot, dry soil where it can get the best of competitors by going dormant during drought, then rebounding after rain returns. Valley fever cases tend to spike after wet winters that follow droughts, says Tom Chiller, chief of the mycotic diseases branch at CDC. And the U.S. West is becoming demonstrably hotter, drier, and more prone to weather extremes.

In the future, climate change “is definitely going to affect Valley fever numbers,” Chiller says. That means the pressure is on to understand where *Coccidioides* hides, how it moves, and how climate change is likely to affect its range—as well as to develop better ways to detect and prevent infections. Although antifungal medications can cure the disease or at least keep it in check, there is no approved vaccine.

IT’S A MID-OCTOBER MORNING, and Jennifer Head, an infectious disease modeler at the University of Michigan, stands near the dry bed of the Kern River in Bakersfield. Clutching a sun hat, she surveys an expansive, uneven landscape of sand, dirt, and burnt yellow grass. A stone’s throw away, on a paved path that tracks the riverbed, curious cyclists and walkers occasionally crane their necks.

“We just need a place where the drones can launch off, that’s flat and clear,” she says. “I feel like this will work.”



Marc Evans, 75, suffered heart failure soon after he developed Valley fever in 2021.

It’s 24°C and will reach 33°C by afternoon. It hasn’t rained in 5 months. Kern County, home to Bakersfield, has the highest incidence of Valley fever in the country. Head has come to test the air and soil for signs of the fungus—and seek clues to why it is spreading.

Cocci’s ephemeral spores, which can appear in a location one day and be gone a few days later, are devilishly difficult to find. That has made defining the exact range of the fungus a challenge, and scientists have mostly used Valley fever case counts to infer its presence. They now suspect it occupies a laundry list of western and southwestern states outside its classic home states of Arizona and California. It extends as far north as Washington state and across the Southwest to Texas, and it’s present in scattered, semiarid parts of Mexico and Central and South America. Unfortunately, some

In fact, Marc’s Valley fever was improving. But, likely weakened by the stress of his bout with cocci pneumonia, his left ventricle, the workhorse of the heart, was failing to pump remotely enough blood to supply his vital organs. When Arlene was allowed into her husband’s room in the intensive care unit, it was a “chaos” of nurses, she says. “All he kept telling them was, ‘I don’t want a catheter.’ I’m like, ‘You got bigger things to worry about than a catheter.’”

The threat of *Coccidioides* isn’t new. The disease it causes was first described by Argentinean medical student Alejandro Posadas in 1892. Valley fever is familiar in the Southwest, where it has infected wildland firefighters; carrot, beet, and radish pickers; solar power farm builders; and cast and crew members on a Ventura County film set.

Most don’t get as sick as Marc did. In the roughly 40% who get symptoms at all, the fungal spores, nourished in the warm,

endemic states—Texas is exhibit A—don't require physicians to report cases to CDC, which makes it impossible for the agency to keep an accurate tally.

Occasionally, serendipity helps reveal the fungus. In 2010, a boy with no recent travel history was diagnosed with Valley fever after crashing his all-terrain vehicle and wounding his knee at a public park in eastern Washington state, far from its known habitat. Enterprising local public health workers and Chiller's CDC lab identified a strain of cocci in the local soil identical to the one that infected the boy. The discovery “dramatically changed the map,” Chiller says.

Dog infections, which veterinarians are alert for, also suggest the fungus is extending its range. (The animals are likely sickened at higher rates, and more severely,

than humans.) An analysis of 835,000 canine blood tests in the United States from 2012–22 found 37.6% were positive, and the geographic spread was unmistakable: 2.4% of U.S. counties had positive canine tests in 2012; by 2022, 12.4% of counties reported positive tests. “There are clearly gaps [in] our human knowledge that dog data has really brought out,” says small-animal internist Jane Sykes at the University of California (UC), Davis, whose team published the results last week in a preprint. (The work has not yet undergone peer review.)

Modeling studies that incorporate climate change suggest the spread will continue. A 2019 analysis led by climate scientist Morgan Gorris at Los Alamos National Laboratory found that southwestern counties with high Valley fever caseloads had higher av-

erage temperatures and lower average precipitation than other counties—conditions climate change will exacerbate. The study predicted that in a high-warming scenario, the fungus would colonize almost the entire western U.S. by 2100 and cases would grow by 50%. “The habitat suitable for *Coccidioides* is going to spread,” Head predicts. “But we don't really know how the spores will get there.”

One idea is that they hitch a ride on wildfire smoke. “When wildfires burn, they mobilize massive quantities of dirt,” Head says—along with a plethora of diverse, living bacteria and fungi. No studies to date have found cocci in smoke, but a 2023 analysis of data from 22 California hospitals, 18 of them in or around major coastal cities, found a 20% increase in admissions for Valley fever following wildfire smoke exposure.

At their first study site of the day in Bakersfield, Head and her colleagues are refining their drone sampling techniques before they take on the harsh environment of controlled burns. In a proof of principle, they have already gathered two positive samples during 41 drone flights above the cocci-laden grasslands of the Carrizo Plain west of Bakersfield—the first cocci spores a drone has ever captured.

Now, they are on the hunt for more clues. The first of two drones lifts off over the brown scrubland. It will pull air onto two filters at a human breathing rate of 7.5 liters per minute, for 20 minutes. In the lab, the team will analyze the filters' catch for cocci DNA. The drones also gather data on temperature, humidity, and particulate matter every 8 seconds—shedding light, the scientists hope, on which combination of conditions are friendliest to airborne cocci spores. Because so little is known about cocci's airborne spread, “Anything we find is useful,” says Environmental Protection Agency microbiologist James Markwiese, who leads the drone project.

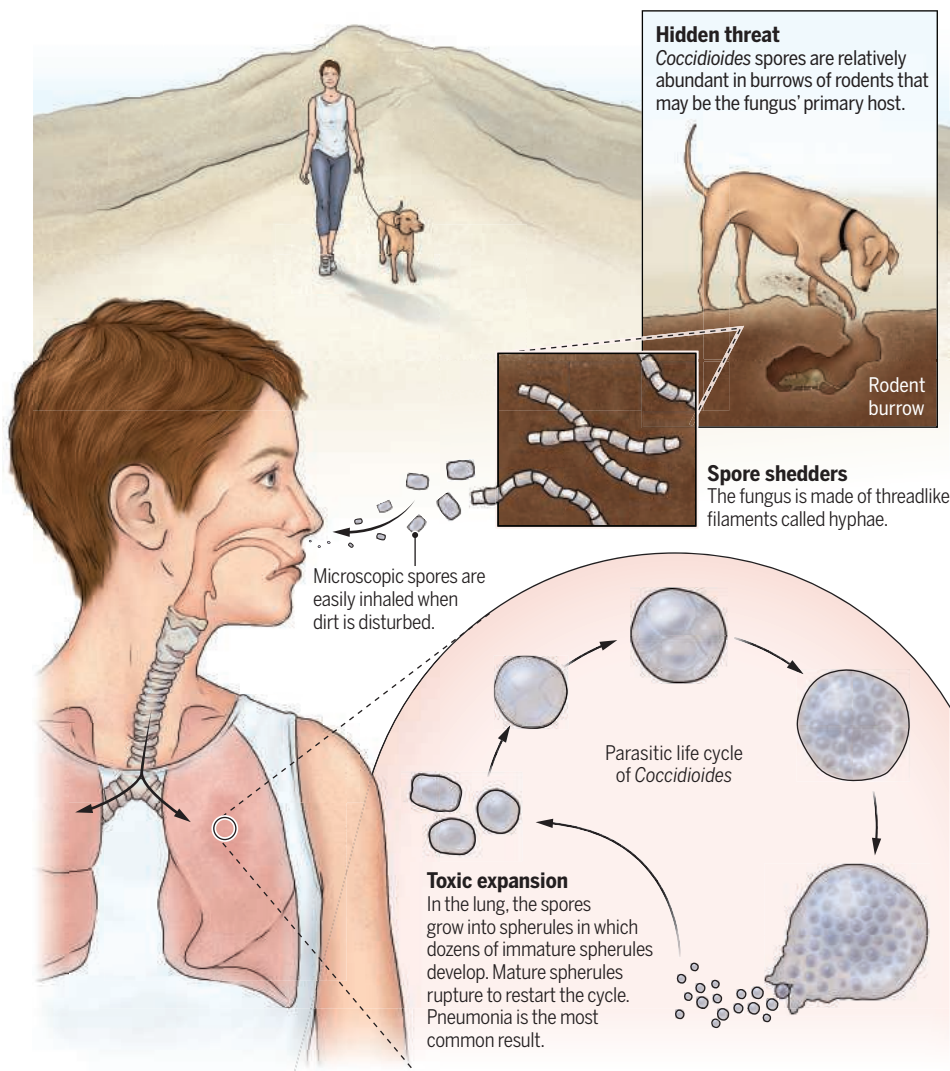
The sampling in Bakersfield could also help researchers tackle another mystery. “We don't even know the dose of spores it takes for somebody to get ill,” Head says. “We don't really know if these spores are somewhat ubiquitous and you just need a higher dose for infection to happen ... or if it takes only a few spores.”

To find out how prevalent the spores are, the team will compare the probability of finding them at altitudes of 12 and 20 meters, on stationary air filters set on tripods at chest height, and in the soil.

Such sampling can also probe the source of the spores people inhale. Recent work led by computational biologist Robert Wagner, a former postdoc in the lab of mycologist John Taylor at UC Berkeley, found that in

Stirring up trouble

Valley fever is caused by the soil-dwelling fungus *Coccidioides*, which outcompetes other microbes in dry, hot conditions and is endemic in the U.S. Southwest. The fungus has recently been found as far north as Washington state, and modelers predict further spread. Dogs and humans are both at risk.





Infectious disease modeler Jennifer Head collects soil near rodent burrows in Bakersfield, California, in October 2024. Soil and air sampling may unravel how humans are exposed to *Coccidioides* spores and how they travel.

the San Joaquin Valley, the spores in the air are genetically quite different from those in the soil. The difference suggests the spores captured at altitude may have blown in from afar instead of being kicked up from nearby ground.

TO FIND SPORES in the soil, Head digs where they are most likely to be. She dons an N95 mask and walks out onto the bumpy, dusty turf. Pulling out a silver instrument resembling an undersize soup ladle, she crouches beside a cantaloupe-size hole on a small rise, pointing out bits of rodent-clipped vegetation and tracks near its mouth that indicate this burrow is active. She sprays the ladle with 70% ethanol and scoops up dirt from near the mouth of the burrow and then from within it, depositing it in plastic tubes.

Typically, cocci “is just such a booger to find,” Markwiese says, looking on.

Rodent burrows are a good bet. It’s been known for decades that diverse rodent species harbor cocci, mostly without showing symptoms, and the soil in rodent burrows is much more likely to test positive than random soil samples, where spores are vanishingly rare. The wind can sweep up spores in the loose dirt rodents leave near the mouths of burrows and carry them to nearby humans.

The rodents could also be spreading the spores from place to place. A recent study

by ecologist Rebecca Rowe’s group at the University of New Hampshire found that in general, spores of soil-dwelling fungi were more likely to catch a ride on small mammals than on the breeze. Although the study did not include *Coccidioides*, its structure “is not quite as aerodynamic as other fungi that are shooting their spores off into space,” Head notes.

A rodent connection could explain why Valley fever cases tend to surge after droughts end, when rodent numbers also burgeon. Head and her group have helped nail down the drought link, crunching data on cases in California from 2000 to 2020 to show that droughts temporarily suppressed case numbers, though they more than bounced back immediately after droughts ended. The study, published in 2022, made Valley fever the first infectious disease causally linked to drought.

The standard explanation for that pattern is the long-held “grow and blow” hypothesis: Cocci mycelia, networks of spore-shedding filaments called hyphae that inhabit soil, go dormant during drought. But when a wet winter follows, they proliferate. After such a winter, dry and hot weather (think Central Valley summers) creates dusty conditions that trigger the hyphae to form spores that can be kicked or blown up from the soil. Recent California case trends—huge numbers of cases after the last two wet winters broke a yearslong

drought—have offered dramatic support for this hypothesis (see graphic, p. 244).

But rodent populations may be as important as weather. The “rodent hypothesis,” first published in 1942, holds that, rather than being accidental hosts of *Coccidioides*, burrowing rodents are primary hosts, harboring the fungus, alive but inactive, in their lungs. When the rodent dies, the theory goes, the fungus morphs into the mycelial form and feeds on the carcass to grow and produce spores.

Taylor and soil microbiologist Bridget Barker at Northern Arizona University (NAU) found support for that scenario in cocci’s own DNA. With fellow microbiologist Thomas Sharpton at Oregon State University and colleagues, they showed *Coccidioides* has over time shed genes useful for digesting plants and multiplied genes for digesting animal protein.

“We know that rodent populations can bottom out during drought and then skyrocket afterwards,” Head notes. If dead infected rodents really introduce spores into soil, drought related die-offs could increase that spore supply. And during a population rebound, nutrients from the shedding of skin, hair, and excreta into soil could nourish fungal growth.

If the rodent hypothesis is right, masking or avoidance of burrow-laden areas could help lower the risk for people living in *Coc-*

coccidioides hot spots. (Taylor was employed as an expert witness in a trial on behalf of California Department of Transportation construction workers who sued the agency after they were infected with cocci in an area covered with rodent burrows. He found that burrows and surface soil where they had excavated were positive for cocci, and the workers won a \$12 million award from the transit agency.) It also means modelers trying to predict the effects of climate change on cocci's range should incorporate not only its direct effects on the fungus, but also how it will drive rodents to travel in search of vegetation, or die off for lack of it. "Understanding if rodents, as they migrate, are capable of introducing *Coccidioides* to a new location will be important to understanding its geographic expansion," Head says.

As Head finishes her soil sampling, the first drone lands. The researchers have 17 sites to hit in 5 days, so they will soon pack up the gear for the drive to the next one; they'll keep launching drones and scooping dirt from rodent burrows until dusk.

BAKERSFIELD NATIVE BRYNN CARRIGAN, 41, director of Kern County Public Health, occasionally ran on the paved path along the Kern River as she trained for the Bakersfield Marathon early in 2024. With the 7 April race in her sights, she recalls, "I was running outside all over the place."

In March, Carrigan, who had never had a health problem beyond an occasional, brief cold, developed a cough and sore throat that lingered for more than 2 weeks. It was followed by a fleeting, irritating rash on her legs. In early April, the headaches started.

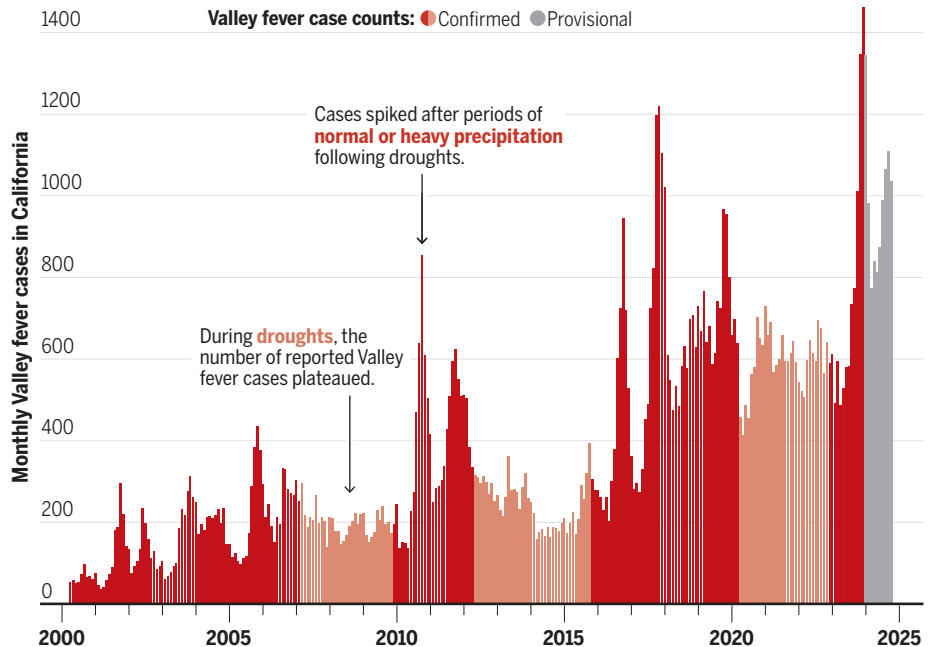
They were so bad she skipped the marathon. Instead, she lay flat in her room with intense photophobia, a blanket pulled over her head. Relentless nausea and vomiting set in. During a series of doctor's office, urgent care, and ER visits, x-rays and CT scans of her lungs showed nothing. Carrigan was diagnosed first with tension headaches—massages were prescribed—and then with dehydration. "I was given some IV fluids and sent home with pain meds. And I knew that wasn't right."

She returned to the ER where "finally, somebody decided to do a spinal tap on me," Carrigan recalls. The fluid bathing her spine should have been clear and colorless; its cloudy, yellowish appearance suggested an infection. On 3 May, she was diagnosed with *Coccidioides* meningitis. Mortality rates in untreated meningitis exceed 90%.

It had not occurred to Carrigan, or any but her latest doctors, that she might have Valley fever. Only looking back did she, and they, connect the dots.

Days when the rains came

In California, hot, dry summers following wet winters—especially after multiyear droughts—have sent Valley fever cases soaring, amplifying typical fall peaks. The "grow and blow" hypothesis holds that rains cause the fungus to proliferate underground. Then, the heat and dryness that follow enhance the release of spores into soil, and their airborne dispersal.



In the hottest spot for the disease in the country, "We're sort of just used to hearing [about] it and maybe not taking it as seriously as we should," she says. "I'm a prime example of why we should not become complacent."

The average delay between seeking medical care and being diagnosed with Valley fever is 38 days. More than half of patients see a provider three or more times before anyone thinks of testing them. Because results can take days to weeks to come back, patients may disappear in the meantime. What's more, the disease is a great masquerader. It can appear to be a skin infection called cellulitis and has regularly been misdiagnosed as tuberculosis. One patient

in Phoenix, 69-year-old Lori Doemland, had no symptoms apart from loss of balance and muscle spasms when she got the disease in 2021. A doctor suggested to her husband that she see a psychiatrist. "My husband was so pissed," she recalls.

Weeks wasted on diagnosis can do great damage if the fungus turns aggressive and escapes the lungs. Although immunocompromised and pregnant people are at much higher risk of developing disseminated disease because their immune systems are operating at a lower ebb, most people who meet this fate don't fall into either category, and scientists are probing why. One team, led by Steve Holland at the National Institute of Allergy and Infectious Diseases, found that people with disseminated disease harbor genetic variations that hobble their immune response to the fungus.

The path back has not been easy for Carrigan. The high-dose antifungal drug fluconazole made her hair and eyelashes fall out, and dried her mouth badly, making it hard to get through public speeches, a big part of her job. Then a relapse landed her back in the hospital in September. She will need to stay on medication for the rest of her life.

The knowledge that her body harbors an infection that would kill her if not for the drugs—and that the chronic stress on her immune system may affect her ability to fight off other microbes—"leaves me feeling vulnerable," she says.



Brynn Carrigan, Kern County director of public health, developed meningitis after inhaling *Coccidioides* spores last year.

**“I loved his smile.
He hasn’t
smiled since
Valley fever.”**

Maria Alcantara



Maria Alcantara (left) with her partner Jose Sanchez, who needs monthly injections into the fluid around his brain to control his *Coccidioides* meningitis.

Still, Carrigan is extremely grateful, and recognizes things could have gone worse—as they did for Jose Sanchez, 68, an architect who after immigrating from Mexico worked for 10 years as a plumber on dusty California construction sites. Months after his diagnosis with *Coccidioides* meningitis in 2019, inflammation from the fungal infection prompted a stroke that immobilized his right side. He now visits the Valley Fever Institute every 4 weeks to have amphotericin B, a powerful antifungal, injected directly into the fluid around his brain via a reservoir implanted under the skin of his scalp.

“Hopefully one day there is something better,” a teary Sanchez said through a translator as he prepared to be sedated for a treatment in October 2024.

Maria Alcantara, Sanchez’s partner of 25 years, left her job in a plastics factory to care for him. But they have burned through their savings and she recently went back to work in a hotel. “I loved his smile,” Alcantara says. “He hasn’t smiled since Valley fever.”

Barker of NAU hopes to prevent outcomes like Sanchez’s with a vaccine that might someday protect at-risk populations, beginning with the new residents flooding the Southwest, including “snowbirds” retiring to the area from colder northern states. With colleagues led by microbiologist Deborah Fuller at the University of Washington, she has used bioinformatics to identify three

Coccidioides antigens that generate a strong immune response. They have shown a prototype vaccine containing DNA from the three antigens can shut down the fungus’ dissemination in mice and are preparing to test it in macaques. Further development, they hope, will lead to a messenger RNA (mRNA) vaccine for use in humans.

Another vaccine group is further along. Physician John Galgiani, plant scientist Marc Orbach, and veterinarian Lisa Shubitz of the University of Arizona (UA) have developed a dog vaccine, based on a live, weakened version of the fungus, that is working its way through the U.S. Department of Agriculture approval process. Barker hopes it succeeds. In 2015, she lost her heeler-pitbull-Dalmatian mix, Victor, to Valley fever, which ablated his lungs and riddled his bones. “Protecting dogs is a huge piece, right? I mean, because we love them. We love them and they suffer.”

Now, the UA team is working with two biotech companies to develop the vaccine for human use. “It seems to be pretty effective, but you’re also injecting a live fungus,” Barker says, which could potentially lead to unwanted side effects. She hopes her group’s planned mRNA vaccine can sidestep such risks.

MARC EVANS SURVIVED that terrifying August night in 2021 thanks to the medical team at Mayo, which kept him alive long

enough for ramped-up antifungals, antibiotics, and heart medications to bring him back from the brink. He was discharged, on oral antifungal medication, after nearly 2 weeks. But his heart was far from better. “I remember very vividly walking the dog one morning and thinking, ‘I’m dying,’” Evans recalls.

He now has an implanted mechanical pump to keep his damaged left ventricle functioning. Often, this is a temporary fix while patients await a heart transplant. But after months of review, a hospital committee turned Evans down for a transplant. The immunosuppressive drugs he would need to take to keep his body from rejecting the new heart could all too easily allow his latent *Coccidioides* infection to flare up, with serious or fatal consequences.

Evans rarely walks his dog anymore. He struggled mightily to attend his son’s 2023 wedding. And when he at last ventured to fly again in the fall of 2024, security staff at the Los Angeles International Airport detained him for 45 minutes, suspicious of the wire protruding from his chest and the attached battery pack.

For others living in the expanding zone where *Coccidioides* is a threat, he warns: “Be mindful of that nagging shortness of breath that doesn’t go away. There are a lot of people out there that just sort of power through. Don’t ignore the symptoms. This disease is a slippery slope.” ■



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